

P-value

P-value

The P-value is the probability of observing data as extreme as that observed, **if the null hypothesis is true**, and **under infinite replications** of the experiment/study.

American Statistical Association (ASA)'s Statement on p-values [March 2016]: <http://www.amstat.org/asa/files/pdfs/P-ValueStatement.pdf>

- Wasserstein R.L., Lazar N.A. (2016). The ASA's Statement of p-values: Context, Process, and Purpose. *The American Statistician*, 70(2), 129-133.
- Dorey F. (2010). In brief: The P-value: What is it and what does it tell you? *Clin Orthop Relat Res*, 468(8), 2297-2298.

Example 1.13

Suppose that a z-statistic of $z = 4.6$ is calculated from a certain data set. It follows that

$$\text{P-value} \approx \frac{1}{1,000,000}.$$

In words, this means that:

If H_0 were true, then the chance of observing data at least as extreme as what was actually observed is 1 in 1 million.

Hence, believing H_0 requires us to believe that the observed data arose as a 1 in a million chance occurrence. This is not a plausible explanation for the data so we reject H_0 .

The smaller p-value, the stronger evidence to reject H_0 .

Example 1.14

Suppose $y_1, y_2, \dots, y_n$ are i.i.d. $N(\mu, \sigma^2)$ observations with known σ^2 . Consider the null hypothesis

$$H_0: \mu = \mu_0,$$

with test statistic

$$z_{obs} = \frac{\bar{y} - \mu_0}{\sigma/\sqrt{n}}$$

then

$$P\text{-value} = P(|Z| \geq |z_{obs}|)$$

for $Z \sim N(0, 1)$.

$$P\text{-value} = P(Z \geq z_{obs})$$

$$P\text{-value} = P(Z \leq z_{obs})$$

$$H_0: \mu = \mu_0$$

$$\mu \leq \mu_0$$

$$\mu \geq \mu_0$$

$$H_a: \mu \neq \mu_0$$

$$\mu > \mu_0$$

$$\mu < \mu_0$$

$$H_a: \mu \neq \mu_0$$

$$H_a: \mu > \mu_0$$

$$H_a: \mu < \mu_0$$

$$H_0: \mu = \mu_0$$

$$\mu \leq \mu_0$$

$$H_a:$$

$$\mu > \mu_0$$

$$\mu < \mu_0$$

Example 1.14 (cont.)

$$\Phi(-x) = 1 - \Phi(x)$$

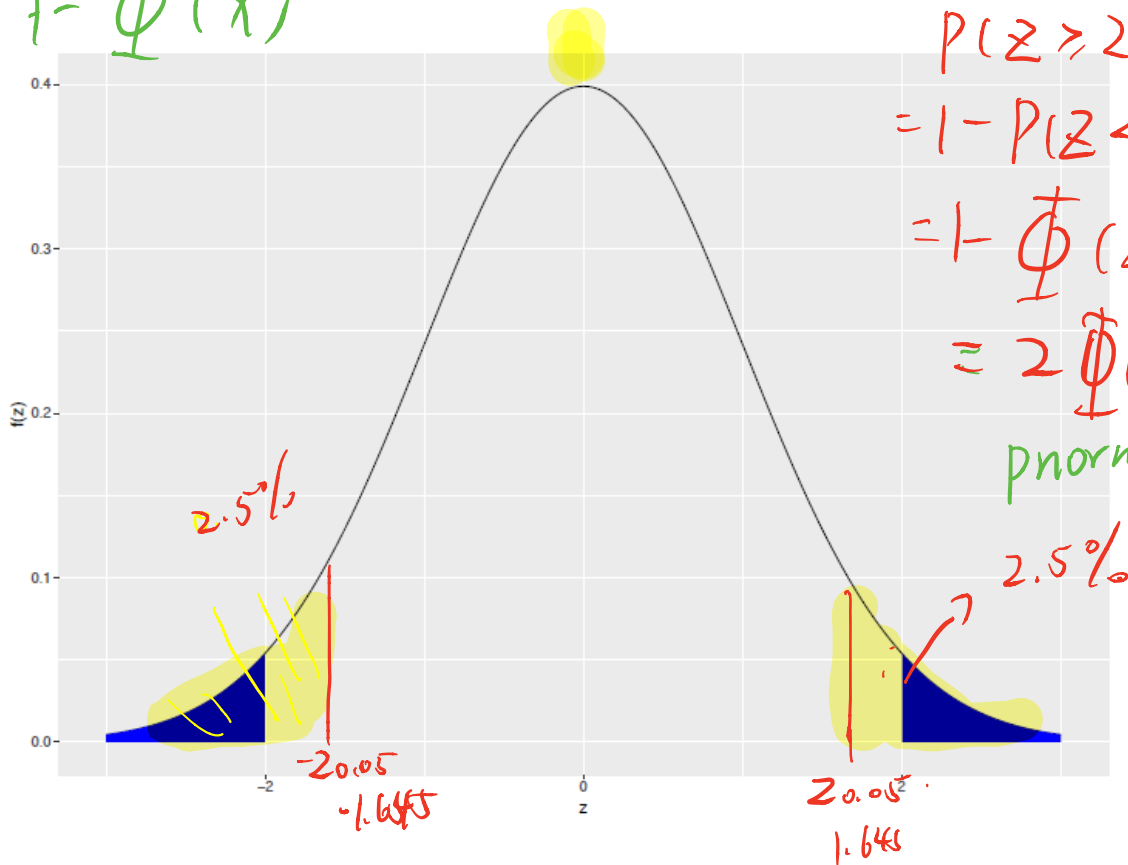
$$\alpha = 0.1$$

$$Z_{0.05} = 1.645$$

$$Z_{0.05} = 2$$

$$\begin{aligned} P(|Z| \geq 2) &= \\ &= P(Z \geq 2) + P(Z \leq -2) \\ &= 1 - P(Z < 2) + P(Z \leq -2) \\ &= 1 - \Phi(2) + \Phi(-2) \\ &= 2\Phi(-2) \end{aligned}$$

$$\text{pnorm}(-2, 0, 1)$$



Two-sided test using P-values

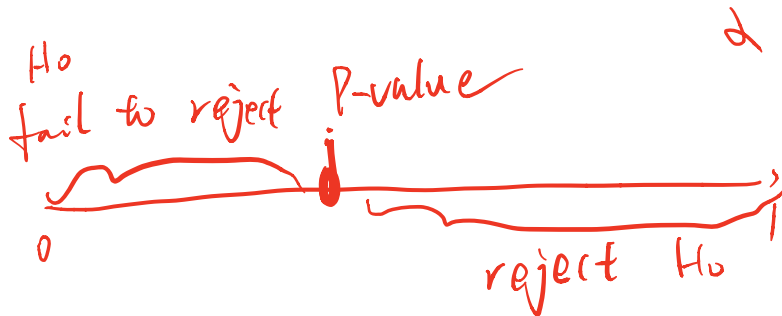
The two-sided hypothesis test can then be formulated equivalently as:

- Reject H_0 if P-value $\leq \alpha$
- Fail to reject H_0 if P-value $> \alpha$

Alternative interpretation of P-values

The P-value is the smallest level of significance α for which the observed data indicate that the null hypothesis should be rejected.

O.S.L. observed significant level



Guidelines

P-value	conclusion
$P\text{-value} > 0.1$	no evidence against H_0
$0.05 < P\text{-value} \leq 0.1$	weak evidence against H_0
$0.01 < P\text{-value} \leq 0.05$	strong evidence against H_0
$P\text{-value} \leq 0.01$	Very strong evidence against H_0

Example 1.15

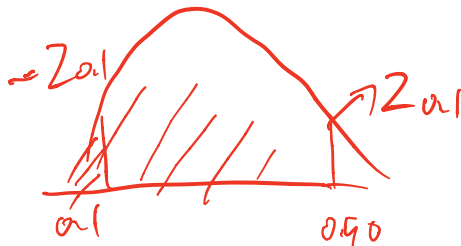


High airline occupancy rates on scheduled flights are essential for profitability. Suppose that a scheduled flight must average at least 60% occupancy to be profitable and that an examination of the occupancy rates for 120 10am flights from Adelaide to Sydney showed mean occupancy rate per flight of 58% and standard deviation 11%.

$\bar{y}$

s

Test to see if sufficient evidence exists to support a claim that the flight is unprofitable. Find the P-value associated with the test. What would you conclude if you wished to implement the test at the $\alpha = 0.10$ level?



$$RR: Z_{obs} < -Z_{\alpha}$$

reject

$$q_{norm}(\alpha, 0.1)$$

Example 1.15 Solution

μ = mean of occupancy
(无利)

$$H_0: \mu \geq 0.6 \text{ (有利)}$$

$$H_a: \mu < 0.6$$

$$Z_{obs} = \frac{\bar{y} - \mu_0}{s/\sqrt{n}} = \frac{0.58 - 0.6}{0.11/\sqrt{120}} = -1.99$$

$$p_{norm}(-1.99, 0, 1)$$

$$P\text{-value} = P(Z < \overset{-1.99}{Z_{obs}}) = P(Z < -1.99) = 0.0233$$

$\therefore P\text{-value} < \alpha$,

$\therefore$ reject H_0

$\therefore$ 无利

Example 1.16



Buyers have claimed that shopping online for a gaming laptop can save a considerable amount – an average of \$900. To test this claim, a random sample of 35 customers who recently brought a gaming laptop online were asked the amount they saved by purchasing online. The mean and standard deviation of this sample were \$885 and \$50 respectively.

- $\bar{y}$ s
- a) Using the usual z-test, at the significance level of $\alpha = 0.01$, is there sufficient evidence to indicate that the average savings differed by \$900?
 - b) Construct a 99% confidence interval for the true average savings. What is our conclusion?
 - c) Find the P-value associated with this test.

Example 1.16 Solution

- a) $H_0: \mu = 900$ $H_a: \mu \neq 900$
 2) test-stat: $Z = \frac{\bar{Y} - \mu_0}{\sigma/\sqrt{n}}$ $\underbrace{\text{under } H_0}_{N(0,1)}$
 3) $\alpha = 0.01$: RR: $\{|Z_{obs}| \geq Z_{0.005} = 2.58\}$

4) $Z_{obs} = \frac{885 - 900}{50/\sqrt{35}} \stackrel{q_{norm}(0.995, 0, 1)}{\approx} -1.77$

$\therefore Z_{obs} \notin RR$ $\therefore$ fail to reject H_0

$\therefore$
 b) 99% CI for μ : $\bar{y} \pm \frac{\sigma}{\sqrt{n}} z_{\frac{\alpha}{2}}$

$= 885 \pm 2.58 \frac{50}{\sqrt{35}}$
 $= (863.195, 906.805)$ — $\therefore 900 \in CI$, fail to reject H_0

c) p-value $= P(|Z| \geq |Z_{obs}|) = P(Z > 1.77) + P(Z < -1.77)$
 $= 2\Phi(-1.77) = 0.0767$ fail to reject H_0

Sample size

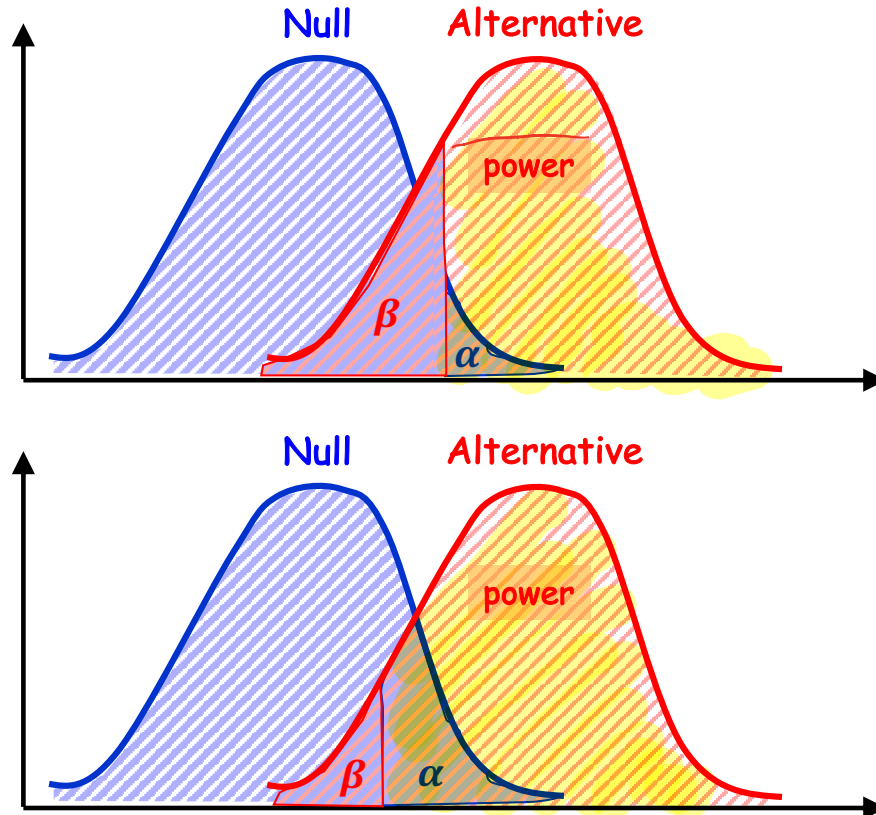
Example 1.17

We plan to conduct a study to compare the means of two equally sized treatment groups using a two-sample z-test.

- a) Assuming a two-sided test, how many patients are required to detect a difference of 1 unit if the standard deviation is 2, using $\alpha = 0.05$ and $\beta = 0.2$?
- b) What is the sample size for a one-sided test with the sample significance level, standard deviation, power and difference?

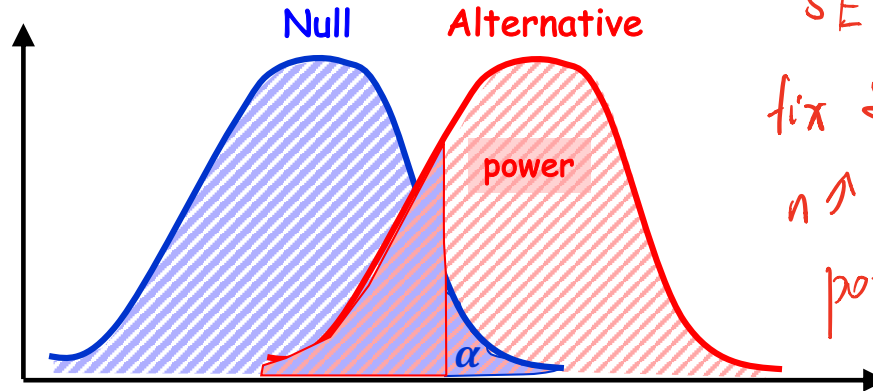
Factors that affect the power of a test

The effect of α on $1 - \beta$



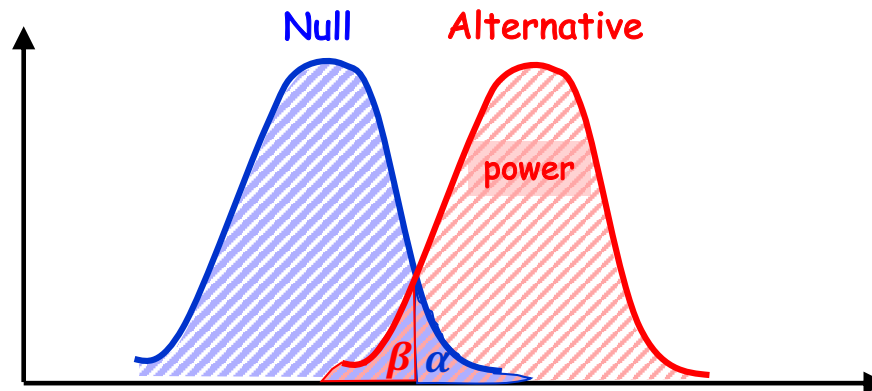
Factors that affect the power of a test

The effect of standard error on $1 - \beta$



$$SE = \frac{\sigma}{\sqrt{n}}$$

fix σ ,
 $n \uparrow \Rightarrow SE \downarrow$
power $\uparrow$, $\beta \downarrow$



Sample size calculations

We can determine the sample size for a given α and β .

Consider testing $H_0: \mu = \mu_0$ against $H_a: \mu > \mu_0$. *one side test*
Given α and β , we want to find the sample size n , for a usual z-test.

We will show that the sample size for this one-sided alternative hypothesis is

$$n = \frac{(Z_\alpha + Z_\beta)^2 \sigma^2}{(\mu_a - \mu_0)^2},$$

two side test

$$n = \frac{(Z_{\frac{\alpha}{2}} + Z_\beta)^2 \sigma^2}{(\mu_a - \mu_0)^2}$$

where β is evaluated when $\mu = \mu_a$ and $\mu_a > \mu_0$.

Sample size calculations (proof)

$$\begin{aligned}
 \alpha &= P(\text{Type I error}) \\
 &= P(\text{reject } H_0 \mid H_0 \text{ true}) \\
 &= P(\bar{Y} > c \mid \mu = \mu_0) \quad Z \sim N(0,1) \\
 &= P\left(\frac{\bar{Y} - \mu_0}{\frac{\sigma}{\sqrt{n}}} > \frac{c - \mu_0}{\frac{\sigma}{\sqrt{n}}} \mid \mu = \mu_0\right) \\
 &= P\left(Z > \frac{c - \mu_0}{\frac{\sigma}{\sqrt{n}}}\right) \quad \text{where } Z \sim N(0,1) \text{ under } H_0 \\
 &= P(Z > Z_\alpha)
 \end{aligned}$$

Therefore, we have

$$Z_\alpha = \frac{c - \mu_0}{\frac{\sigma}{\sqrt{n}}} \quad \text{and} \quad -Z_\beta = \frac{c - \mu_a}{\frac{\sigma}{\sqrt{n}}}$$

Let μ_a = true population mean
 $\mu_a > \mu_0$ under H_a

$$\begin{aligned}
 \beta &= P(\text{Type II error}) \\
 &= P(\bar{Y} \leq c \mid \mu = \mu_a) \\
 &= P\left(\frac{\bar{Y} - \mu_a}{\frac{\sigma}{\sqrt{n}}} \leq \frac{c - \mu_a}{\frac{\sigma}{\sqrt{n}}} \mid \mu = \mu_a\right) \\
 &\quad \leftarrow Z \sim N(0,1) \\
 &= P\left(Z^* \leq \frac{c - \mu_a}{\frac{\sigma}{\sqrt{n}}}\right) \quad \text{where} \\
 &= P\left(Z^* > -\frac{c - \mu_a}{\frac{\sigma}{\sqrt{n}}}\right) \quad Z^* \sim N(0,1) \\
 &= P(Z^* > Z_\beta)
 \end{aligned}$$

We have two equations in two unknowns (c and n).

We proceed by eliminating c first.

Sample size calculations (proof)

$$Z_{\alpha} = \frac{c - \mu_0}{\frac{\sigma}{\sqrt{n}}} \quad \text{and} \quad -Z_{\beta} = \frac{c - \mu_a}{\frac{\sigma}{\sqrt{n}}}$$

$$Z_{\alpha} \frac{\sigma}{\sqrt{n}} = c - \mu_0$$

$$c = \mu_0 + Z_{\alpha} \frac{\sigma}{\sqrt{n}}$$

$$-Z_{\beta} \frac{\sigma}{\sqrt{n}} = c - \mu_a$$

$$c = \mu_a - Z_{\beta} \frac{\sigma}{\sqrt{n}}$$

Equating them: $\mu_0 + Z_{\alpha} \frac{\sigma}{\sqrt{n}} = \mu_a - Z_{\beta} \frac{\sigma}{\sqrt{n}}$

$$Z_{\alpha} \frac{\sigma}{\sqrt{n}} + Z_{\beta} \frac{\sigma}{\sqrt{n}} = \mu_a - \mu_0$$

$$(Z_{\alpha} + Z_{\beta}) \frac{\sigma}{\sqrt{n}} = \mu_a - \mu_0$$

$$\frac{\sigma}{\sqrt{n}} = \frac{\mu_a - \mu_0}{Z_{\alpha} + Z_{\beta}}$$

$$\sqrt{n} = \frac{(Z_{\alpha} + Z_{\beta}) \sigma}{\mu_a - \mu_0}$$

$$n = \frac{\sigma^2 (Z_{\alpha} + Z_{\beta})^2}{(\mu_a - \mu_0)^2}$$

Example 1.18

How many samples do we need to determine the mean of a population if, from previous knowledge we know $\sigma = 3.1$ and we want to test the hypothesis $H_0: \mu = 5$ against $H_1: \mu = 5.5$? We also want $\alpha = 0.01$ and $\beta = 0.05$.

$$Z_{0.01} \approx 2.33$$

$$Z_{0.05} = 1.645$$

$$n = \frac{(Z_\alpha + Z_\beta)^2 \sigma^2}{(\mu_a - \mu_0)^2}$$

$$= \frac{(2.33 + 1.645)^2 3.1^2}{(5.5 - 5)^2}$$

$$= 607.37$$

$$n \approx 608$$

Sample size for comparing two means

Consider testing $H_0: \mu_1 = \mu_2$ against $H_a: \mu_1 \neq \mu_2$.

Given α and β , we want to find the sample size n to detect a difference in mean of size δ .

It can be shown that the sample size required (per group) in this case is

$$n = \frac{2\sigma^2 (Z_{\alpha/2} + Z_{\beta})^2}{\delta^2}.$$

→ Z_{α} for one-side test.

✓ $(\mu_2 - \mu_1)^2$

assume,

$$\sigma_1^2 = \sigma_2^2 = \sigma^2$$

$$n_1 = n_2 = n.$$

Example 1.17 (Revisit)

We plan to conduct a study to compare the means of two equally sized treatment groups using a two-sample z-test.

- a) Assuming a two-sided test, how many patients are required to detect a difference of 1 unit if the standard deviation is 2, using $\alpha = 0.05$ and $\beta = 0.2$?

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- b) What is the sample size for a one-sided test with the sample significance level, standard deviation, power and difference?

$$a) : n = \frac{2 \cdot 6^2 (Z_{\frac{\alpha}{2}} + Z_{\beta})^2}{1^2} = \frac{2 \cdot 4 (1.96 + 0.84)^2}{1^2} = \underline{62.72}$$

$$n = 63$$

Example 1.17 Solution

$$b) \quad h = \frac{2.4 \left(\underset{1.645}{Z_2} + 0.84 \right)^2}{1^2} = 49.2032$$

$$n = 50$$

Sample size for comparing proportion(s)

Consider testing $H_0: p = p_0$ against $H_a: p \neq p_0$.

Given α and β , the sample size n to detect a population proportion within $\pm\delta$ margin of error is

$$n = \frac{(z_{\alpha/2} + z_{\beta})^2 p(1-p)}{\delta^2}.$$

When testing $H_0: p_1 = p_2$ against $H_a: p_1 \neq p_2$, the number of samples n required (per group) to detect a difference in p_1 and p_2 , with a given α and β , is

$$n = \frac{(z_{\alpha/2} + z_{\beta})^2 [p_1(1-p_1) + p_2(1-p_2)]}{(p_1 - p_2)^2}.$$

Example 1.19



A researcher plans to conduct a study on the effectiveness of a new drug treatment compared to a standard behavioural treatment for a particular mental illness. It is known that the standard treatment has a cure rate of 70%. Assume a two-sided test is desired, with significance level of 5% and a power of 90%. How many subjects are required to detect an improvement of 10% in the new drug?

$\beta = 0.1$

$$n = \frac{(2 \cdot 0.025 + 2 \cdot 0.1)^2 (0.7 \times 0.3 + 0.8 \times 0.2)}{(0.7 - 0.8)^2}$$

$$= 388.41$$

$$\approx 389$$

Module 1

